COMPUTATION OF SUPERFLUID STIFFNESS FOR A LATTICE MODEL OF ANYON SUPERCONDUCTIVITY

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Abstract. We propose a detailed methodology for computing the superfluid stiffness and its scaling with respect to related observables in a triangular lattice Hofstadter-Hubbard model using density matrix renormalization group (DMRG) techniques. Our goal is to further characterize the superconducting anyon phase, predicted by Divic et al. to emerge upon doping near an integer quantum Hall (IQH) to chiral spin liquid (CSL) topological critical point [1]. This would be the first study of the superfluid stiffness constant and its critical exponents, important signatures of unconventional superconductivity, in a non-disordered lattice-based model of spin liquids.

1. Introduction

Conventional superconductors are relatively well-understood under Bardeen-Cooper-Schrieffer (BCS) theory and its extensions. However, experimental observations of high-T_c and other unconventional superconductors have made clear that the BCS framework is not sufficient to describe every route to superconductivity [13]. In particular, anyons, quasiparticles that have neither Bose-Einstein nor Fermi-Dirac statistics, have long been proposed as a mechanism for unconventional superconductivity in $2+1$ dimensions or fewer [4, 16]. In field theories, such excitations can occur as a result of Chern-Simons terms in the effective Lagrangian, which induce charge-flux coupling and fractional statistics via the Aharonov-Bohm effect [21]. As these statistics are intrinsically tied to the topological order of the underlying phase of matter, anyon superconductivity represents a distinctly topological mechanism of achieving superconductivity.

Anyon superconductivity has gained renewed interest in light of new experimental progress. In particular, the discovery of the fractional quantum anomalous Hall effect (FQAHE) has demonstrated that anyonic excitations can exist in experimentally-accessible Moiré materials like graphene [5] and twisted bilayer $MoTe_2$ [17], motivating greater study of lattice models of anyon superconductivity [8]. Building on these developments, Divic et al. [1] recently identified one such particularly compelling microscopic model. They demonstrated that under a strong magnetic field and strongly-repulsive particle interactions, a Moiré-inspired triangular lattice Hofstadter-Hubbard model can take on a topological phase transition from an integer quantum Hall (IQH) phase to a chiral spin liquid (CSL) phase [1]. Upon finite doping on either side of the phase transition, they observed that a unique, intrinsically topological anyon superconducting phase emerges. Our project will more rigorously analyze this mechanism through the superfluid stiffness of the system, a quantity central to superconductivity that has never been directly extracted in this model.

1.1. Superconducting anyons and Chern-Simons gauge theory.

Both the CSL and IQH phases are gapped phases. However, the CSL phase supports anyons while the IQH phase does not, so the two phases must have different topological orders. Because of this, any phase transition must close the gap, resulting in a gapless transition. This is a very similar mechanism to the one driving experimentally-observed topological insulators (although there the driver is a phase transition between two symmetry-protected topological phases with the same topological order), where the transition between two gapped insulating phases must be a gapless conducting phase [9].

In order for the gapless phase to be superconducting, it must support the formation of Cooper pairs, pairs of electrons which behave like a superfluid in the material, thereby allowing it to

conduct with zero resistance. Under BCS theory, this occurs because of an emergent phonon-induced attractive interaction [13]. Furthermore, in order for this pairing to happen, the system is traditionally required to obey time-reversal symmetry, as in BCS theory, since the electrons in a Cooper pair are essentially time-reversal pairs [12].

The triangular lattice Hofstadter-Hubbard model proposed by Divic et al. then seems like an unlikely candidate for superconductivity, since instead of attractive interactions, it has strong repulsive interactions. Additionally, it breaks time-reversal symmetry. However, numerical evidence from [1] suggest that the formation of Cooper pairs is favorable in this gapless phase. Therefore, the mechanism behind the formation of Cooper pairs must be entirely different.

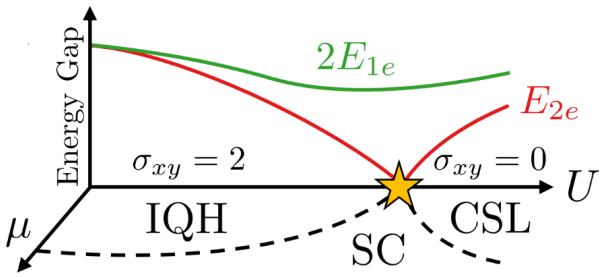


FIGURE 1. Phase diagram from [1]. Between the two topologically distinct gapped IQH and CSL phases, there is a mandatory gapless phase (labeled SC), which hosts Cooper pairs. This diagram is elaborated on further in [1].

Divic et al. argue that this represents anyon superconductivity. The dynamics at the critical point include an effective field theory with scalar field χ coupled to a $U(1)$ gauge field a with the following Lagrangian, which includes a Chern-Simons term of level N:

$$\mathcal{L} = |\mathcal{D}_{A-a}\chi|^2 - V(\chi) + \frac{N}{4\pi} a \wedge da. \tag{1.1}$$

This Chern-Simons term gives an effective mass to the gauge field [14]. However, according to Divic et al., under certain conditions, the Chern-Simons term cancels with other terms, which means a is massless and a superconducting state is supported [1].

It should be mentioned that algebraic approaches to studying anyon interactions are also common. In particular, a category-theoretic approach to anyon superconductivity was recently developed [11]. While useful when studying the phases themselves, in order to verify material properties of this anyon phase transition in matter, including the superfluid stiffness, we must use the more traditional field-theoretic approach.

1.2. **Superfluid stiffness.** The low-energy effective action of a condensed matter system with superconducting phase field θ and mass coupling m contains a term

$$\frac{1}{2}\kappa \int d^d x \frac{\hbar^2}{m} (\nabla\theta)^2 \tag{1.2}$$

where κ is the superfluid stiffness constant [3]. Under a nematic field treatment, this coefficient can be interpreted as the free-energy cost of "twisting" the phase field, and therefore characterizes the stability of the condensate [15].

Whereas BCS theory predicts the stiffness to be a constant dependent only on electron mass and density [20], applying a generic scaling theory to phase transition based superconductivity, as derived in [3] gives the Josephson hyperscaling relation, which states that for a parameter δ such that the characteristic length ξ and frequency Ω scale as $\xi \sim \delta^{-\nu}$ and $\Omega \sim \xi^{-z} \sim \delta^{\nu z}$ near a scale-invariant critical point, we must also have

$$\kappa \sim \delta^{\nu(d+z)-2} \equiv \delta^{-\alpha}. \tag{1.3}$$

This behavior has also been confirmed with renormalization group techniques [10]. Thus, the superfluid stiffness and its scaling have become useful tools for determining anyon superconductivity, since Josephson-like scaling provides evidence of an intrinsically topological critical point.

2. Proposal

We plan to use numerical quantum field theory techniques to analyze the superfluid stiffness constant of the triangular lattice Hofstadter-Hubbard model described in [1]. In particular, we hope to determine if κ exhibits Josephson scaling with the doping y. Furthermore, we plan to measure κ as a function of the interaction strength to hopping amplitude ratio U/t (see definitions of U and t in Section 2.1 below), which, if continuous, would provide stronger numerical evidence for the two superconducting phases created by doping either the IQH or CSL phase being adiabatically connected and hence the same phase, as suggested by the parton construction of Divic et al [1].

2.1. **Methodology.** We plan to simulate the model with purely imaginary nearest-neighbor hopping amplitudes, where each elementary triangle on the lattice has an orbital magnetic flux of $\Phi_\triangle = \pi/2$, as described by Divic et al. [1]. For a given on-site repulsive Hubbard interaction strength $U > 0$ and hopping amplitude $i\tau_{ij} = -t < 0$, the full Hamiltonian, which includes single-particle terms and an interaction term, is

$$H = \sum_{\langle ij\rangle,\sigma} (i\tau_{ij}) \left(c^\dagger_{i\sigma} c_{j\sigma} - c^\dagger_{j\sigma} c_{i\sigma} \right) + U \sum_i \left(n_{i\uparrow} - \tfrac{1}{2} \right) \left(n_{i\downarrow} - \tfrac{1}{2} \right), \tag{2.1}$$

where $c^\dagger_{i\sigma}$ creates a fermion of spin $\sigma \in \{\uparrow, \downarrow\}$ on site i, $n_{i\sigma} = c^\dagger_{i\sigma} c_{i\sigma}$ is the number operator [1]. The nearest-neighbor hopping amplitudes τ_{ij} are defined with an oriented sign convention, so that the product of phases accumulated around an elementary triangle equals $\pi/2$ as desired.

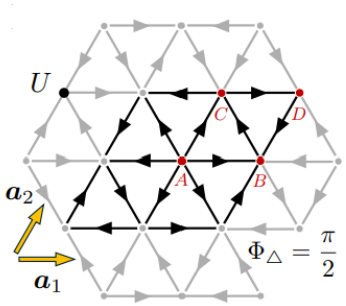


FIGURE 2. The lattice model from [1]. Each triangle has a $\frac{\pi}{2}$ flux.

We plan to extract the low-energy dynamics of this model using infinite density matrix renormalization group (iDRMG) techniques, which are variational methods standard in the literature for microscopic Hamiltonian simulations on a lattice [18]. Following this procedure, we choose the geometry of our system to be an infinite cylindrical lattice in which one boundary is periodic while the other is open, since this topology provides numerous advantages for numerical simulation [18] with relatively good computational efficiency at larger lattice sizes [19].

2.1.1. *Stiffness discretization.* Since our study is performed on a lattice, we must reformulate the definition of the stiffness constant so that it is measurable in a discrete system. We employ the approach found in [2], in which we impose a twisted boundary condition consisting of a phase winding of θ between each site, corresponding to a total winding of θL, where L is the finite circumference of the cylinder in terms of the number of nodes. We can compute the energy per unit length ϵ as a function of this twist θ, and compute the stiffness by

$$\kappa = \frac{1}{L}\left(\frac{d^2}{d\theta^2}\varepsilon(\theta)\right)|_{\theta=0} \tag{2.2}$$

Thus, κ can be determined using this second-derivative definition by measuring the ground-state energy per length $\varepsilon(\theta)$ of the lattice across several small twisted boundary conditions (i.e. $\theta = 0, \delta\theta, 2\delta\theta, ...$).

2.1.2. *Extraction of the dynamical exponent.* As suggested in Equation 1.3, probing Josephson scaling involves the extraction of the critical exponent ν, which requires knowledge of the dynamical exponent z. Although the parton-derived field theory in Equation 1.1 posited by Divic et al. implies $z = 1$, it is unknown how gauge transformations beyond their mean-field approximation might renormalize z [1]. Thus these calculations, also possible through iDRMG techniques, would be necessary despite the extra computational burden.

3. Analyses and Considerations

It should be noted that the scaling of the superfluid stiffness constant has previously been used as a probe for anyon superconductivity in doped FQAH states. Notably, Nosov et al. in [7] extract the dependence of κ on polarizability in a continuous effective field theory of doped FQAH anyons, allowing them to demonstrate Josephson scaling with reference to a topological universality class. However, to our knowledge, these scaling relations have not been extracted for systems with clean (i.e. disorderless) phase transitions, which may represent different topological universality classes and hold different symmetries. In particular, while the disordered system studied by Nosov et al. involves $z = 2$ diffusive dynamics [7], a clean system such as that proposed by Divic et al. should be Lorentz-invariant under $z = 1$ [1].

Moreover, the scaling of the superfluid stiffness constant has never been calculated for a lattice Hofstadter-Hubbard model such as that described by Divic et al.; the closest that previous work has come is computing the critical exponents of a charge density wave (CDW) order parameter indicative of the phase transition [8]. Therefore, our approach would not only analyze a more directly-related signature of superconductivity in the Hofstadter-Hubbard model, but also establish if the lattice Hamiltonian indeed reproduces the continous scaling in Equation (1.3), as has long been conjectured [1].

Finally, probing the stiffness would also allow us to not only analyze properties of the superconducting anyon phase, but also properties of the transition from which it emerges. It is known that phase-transition-driven superconductivity in $2+1$ dimensions is directly related to κ via the Berenzinskii-Kosterlitz-Thouless (BKT) transition [6], which is temperature-based. Thus, knowing the stiffness would also enable characterization of this critical temperature, even with κ only computed at zero temperature.

Moreover, it is also possible to describe κ at finite temperature. The core of the outline above will help explore the phase transition at zero temperature, which is what Divic et al. describe [1]. However, the use of computer simulation allows us to probe the system at finite temperature, which would be intractable by hand. This should help align our results better with the results observed in real-world materials like graphene and $MoTe_2$.

3.1. **Potential challenges.** The novelty of our proposed methodology presents challenges in numerical simulation, and we must take procedural steps to address these. As discussed in subsection 2.1, we had to discretize our definition of lattice stiffness, and must continue to ensure that no unexpected emergent effects occur due to finite lattice size. Additionally, due to the finiteness of our simulation, we must impose sensible boundary conditions. Although anyons must always be on a 2-dimensional manifold, different boundary conditions may result in different topologies, which may have measurable consequences. We can take this as an opportunity to explore how the topology of the system impacts the dynamics by changing the boundary conditions of the simulation in different trials and using that data to control for topological effects.

Getting precise measurements of the stiffness and particularly the dynamical exponent would require extensive computational power, even with the efficiency improvements brought by iDRMG [18], which would require additional resources. However, with these, we would be able to guarantee results that are not only accurate but useful.

It is also worth noting that, more fundamentally, we are only measuring signals of superconductivity, and not superconductivity itself. While necessary to make the simulation tractable, and even though measurements of the stiffness would provide strong evidence for properties of the superconducting phase, we would not be simulating the superconductor directly.

4. Conclusion

By extending the numerical techniques used by Divic et al. to probing the superfluid stiffness, we hope to more closely study the unusual phase transition presented in their work. The advantage of our proposal is that, in any case, it would provide novel contributions to the theory of anyon superconductivity. If the dependence of the stiffness κ on the interaction strength U is continuous, we have numerical evidence that the IQH-CSL transition in this model forms a single phase. Otherwise, we have found evidence against the claim of anyon superconductivity made by Divic et al. Similarly, if the superfluid stiffness obeys Josephson scaling, we have further evidence that the IQH-CSL transition demonstrates anyon-derived superconductivity. Otherwise, our results enhance our understanding of the diverse ways superconductivity can emerge beyond supposedly universal properties such as Josephson scaling.

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